Question			Answer	Marks	AO	Guidance
13	(a)		H_0 $p = 0.494$ H_1 $p < 0.494$	**B1**	**1.1**	allow equivalent in words; do not allow percentages
			p is the **probability** that a person **(**aged 16 or over from England or Wales selected at random) is married or in a civil partnership	**B1** **[2]**	**2.5**	or p is the **proportion** of people (aged 16 or over from England or Wales selected at random) who are married or in a civil partnership; must mention **either** married **or** in a civil partnership **B1B1** for other symbol instead of p used if correctly defined
13	(b)		use of $X \sim B(78,0.494)$ **soi**	**M1**	**3.3**	for calculation of a value for $P(X \leq k)$, ignore slip in notation **M0** for attempt at Normal approximation
			$(P(X \leq 27) =) 0.0059 \ldots$ and$(P(X \leq 28) =) 0.011 \ldots$ seen	**M1**	**3.4**	may see invBinom(0.01,78,0.494) = 28 for **M1M1;**
			$[0 \leq] X \leq 27$ or $[0 \leq] X < 28$ **oe**	**A1** **[3]**	**1.1**	allow critical region < 28 **oe;** if **M0M0** allow **SC1** for correct critical region with no working or insufficient working
13	(c)		26 correctly compared with their 27 or their 28	**M1**	**3.4**	allow eg 26 is in critical region
			reject H_0 **or** accept H_1 **or** significant	**A1**	**1.1**	from correct critical region; condone from $0 < X \leq 27$ **oe**
			there is **sufficient evidence** at the 1% level to **suggest** that the **proportion** of people aged **16 or over** who are **married or in a civil partnership** is **lower** in 2024 than in 2022	**A1** **[3]**	**2.2b**	dependent on award of previous two marks and correct critical region; allow percentage/probability but not "number" do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; if **M0** allow **SC1** for $\mathbf{P(X \leq 26) = 0.003 < 0.01}$, so **reject H_0** and conclude there is **sufficient evidence** at the 1% level to **suggest** that the **proportion** of people aged **16 or over** who are **married or in a civil partnership** is **lower** in 2024 than in 2022

Question			Answer	Marks	AO	Guidance
13	(d)		0.01 **oe**	**B1** **[1]**	**1.2**	allow [P($X \leq 27$) =] 0.005 946….